SCHOOL ENROLMENT, ATTENDANCE AND STUDENT RETENTION ANALYSIS

SQL Data Analysis Project Report

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Dataset note: This report uses a school practice dataset. Results are practice findings, not official statistics for a real school.

1. Introduction

This project uses SQL in MySQL Workbench to analyse enrolment, attendance, student movements and data quality. The purpose is to identify patterns that may need further investigation. Because the dataset is for practice, results should be checked against authorised school records before being used for real decisions.

2. Analysis and Findings

Question 1: What is the total student enrolment?

Objective: Count distinct students enrolled in Term 1 (T1).

SQL query:

```
SELECT COUNT(DISTINCT student_id) AS total_enrolment
FROM enrolments
WHERE term_id = 'T1';
```

Result: 119 students.

Interpretation: Each student is counted once for T1.

Question 2: How many students are enrolled in each class?

Objective: Count distinct students in each class for T1.

SQL query:

```
SELECT c.class_name,
    COUNT(DISTINCT e.student_id) AS total_students
FROM enrolments e
JOIN classes c ON e.class_id = c.class_id
WHERE e.term_id = 'T1'
GROUP BY c.class_id, c.class_name
ORDER BY c.class_name;
```

Result: Grade 7 East: 13; Grade 7 West: 19; Grade 8 East: 23; Grade 8 West: 22; Grade 9 East: 21; Grade 9 West: 21. Total: 119.

Interpretation: Grade 8 East has the largest enrolment (23); Grade 7 East has the smallest (13).

Question 3: What is the overall attendance rate?

Objective: Calculate Present records as a percentage of all attendance records.

SQL query:

```
SELECT ROUND(
  SUM(CASE WHEN attendance_status = 'Present' THEN 1 ELSE 0 END)
  * 100.0 / COUNT(*), 2
) AS attendance_rate_percent
FROM attendance;
```

Result: 76.29%: 3,555 Present records out of 4,660 attendance records.

Interpretation: All attendance rows are in the denominator; Excused and Late are not counted as Present.

Question 4: Which classes have the lowest attendance rate?

Objective: Calculate and sort each class's percentage of records marked Present.

SQL query:

```
SELECT c.class_name,
    ROUND(SUM(CASE WHEN a.attendance_status = 'Present' THEN 1 ELSE 0 END)
    * 100.0 / COUNT(*), 2) AS attendance_rate_percent
FROM attendance a
JOIN classes c ON a.class_id = c.class_id
GROUP BY c.class_id, c.class_name
ORDER BY attendance_rate_percent ASC;
```

Result: Grade 8 West: 68.64%; Grade 9 West: 72.07%; Grade 9 East: 78.78%; Grade 8 East: 79.13%; Grade 7 West: 79.29%; Grade 7 East: 82.88%.

Interpretation: Grade 8 West has the lowest attendance rate and is a useful starting point for further investigation.

Question 5: How does attendance change by month?

Objective: Calculate the percentage of Present records for each month.

SQL query:

```
SELECT DATE_FORMAT(attendance_date, '%Y-%m') AS attendance_month,
    ROUND(SUM(CASE WHEN attendance_status = 'Present' THEN 1 ELSE 0 END)
    * 100.0 / COUNT(*), 2) AS attendance_rate_percent
FROM attendance
GROUP BY DATE_FORMAT(attendance_date, '%Y-%m')
ORDER BY attendance_month;
```

Result: Jan 2026: 76.47%; Feb: 77.55%; Mar: 77.67%; Apr: 72.05%; May: 75.44%; Jun: 74.56%; Jul: 77.37%; Aug: 77.19%.

Interpretation: April was lowest (72.05%) and March was highest (77.67%). The reason for April's decline needs investigation.

Question 6: How many students transferred or withdrew?

Objective: Count distinct students by movement type.

SQL query:

```
SELECT movement_type,
    COUNT(DISTINCT student_id) AS total_students
FROM student_movements
GROUP BY movement_type
ORDER BY total_students DESC;
```

Result: 4 transfers and 2 withdrawals; 6 movement records in total.

Interpretation: The practice dataset records more transfers than withdrawals.

Question 7: How does attendance compare between terms?

Objective: Calculate attendance rate separately for each term.

SQL query:

```
SELECT term_id,
    ROUND(SUM(CASE WHEN attendance_status = 'Present' THEN 1 ELSE 0 END)
    * 100.0 / COUNT(*), 2) AS attendance_rate_percent
FROM attendance
GROUP BY term_id
ORDER BY term_id;
```

Result: T1: 76.81%; T2: 75.75%.

Interpretation: Attendance decreased by 1.06 percentage points from T1 to T2.

Question 8: Are there data quality problems?

Objective: Check missing values, duplicate student IDs, and missing attendance details.

SQL query:

```
-- Missing student ID or gender
SELECT * FROM students
WHERE student_id IS NULL OR TRIM(student_id) = ''
  OR gender IS NULL OR TRIM(gender) = '';

-- Duplicate student IDs
SELECT student_id, COUNT(*) AS records
FROM students
GROUP BY student_id
HAVING COUNT(*) > 1;

-- Missing attendance details
SELECT * FROM attendance
WHERE student_id IS NULL OR TRIM(student_id) = ''
  OR attendance_date IS NULL
  OR attendance_status IS NULL OR TRIM(attendance_status) = '';
```

Result: The three checks shown above returned no rows: no missing student ID or gender, no duplicate student IDs, and no missing attendance student ID, date or status.

Interpretation: The fields checked appear consistent in this practice dataset. These checks do not prove that every possible data-quality issue is absent; additional checks may be added when required.

3. Summary of Key Findings

- Term 1 enrolment was 119 students.
- Enrolment across the six classes ranged from 13 to 23 students.
- Overall attendance was 76.29% across the records analysed.
- Grade 8 West had the lowest class attendance rate (68.64%).
- April had the lowest monthly attendance rate (72.05%); March had the highest (77.67%).
- Student movement records showed 4 transfers and 2 withdrawals.
- Attendance declined from 76.81% in T1 to 75.75% in T2.
- The data quality checks described above did not identify issues in the tested fields.

4. Recommendations

- Investigate attendance in Grade 8 West to understand possible reasons for the low rate.
- Review the April attendance decline and compare future monthly trends.
- Monitor attendance by class and term so changes can be identified early.
- Follow school procedures to contact families and support students with repeated absences.
- Keep transfer and withdrawal records complete, including dates and reasons where available.
- Expand future data-quality checks to include invalid attendance labels and unmatched student/class IDs.
- Verify these practice findings against authorised school records before using them for real decisions.

5. Conclusion

This project demonstrates how SQL can summarise enrolment, compare attendance across classes and months, review student movements, and check data quality. The practice dataset shows differences between classes, a low monthly rate in April, and a small decline from T1 to T2. These results can guide further questions but do not establish the causes. Verify all results against official school records before using them for planning.